HIGH AFFINITY ARTIFICAL BINDERS FOR INTEGRATED CHARACTERIZATION OF AN ESSENTIAL MALARIA PROTEIN.

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ABSTRACT

Despite major progress in malaria control, the disease remains a significant global health challenge[1] driven by the emergence of multidrug resistance among parasite species. Addressing this challenge requires the need to identify and characterize essential parasite proteins with therapeutic potential. One promising target is the *Plasmodium falciparum* Ca^{2+}-ATPase 6 (PfATP6), an essential and unique calcium P-type ATPase[2]. The aim of this study is to develop molecular tools to investigate the structure, function and endogenous localization of PfATP6 to improve its potential as an antimalarial target.

Here we present a procedure to immobilize purified PfATP6 solubilized in detergent on Elisa plates and screen a diverse library of artificial proteins (alphareps) by phage display. Iterative rounds of affinity selection led to the identification of two highly specific alpharep variants binding PfATP6 with nanomolar affinities. BLI experiments shows that the interaction is independent of PfATP6 conformation, with both alphareps showing comparable nanomolar affinities across different PfATP6 calcium transport cycle. Additionally, alphareps binding did not interfere with the ATPase activity of PfATP6, supporting their use as suitable structural and functional tools.

Lately, preliminary cryoEM data revealed that both alphareps bind the same specific cytoplasmic domain of PfATP6, in agreement with Alphafold predictions (*figure 1*).

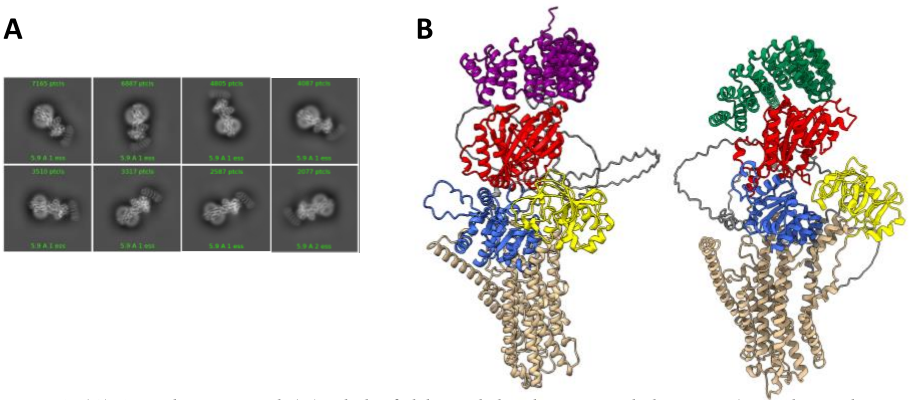

Figure 1:(A) 2D classes and (B) alphafold models showing alphareps (purple and green) binding on specific domain of PfATP6.

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